AN F-MAP OF SHEAVES WITH VALUES IN AN ALGEBRAIC STRUCTURE IS DETERMINED BY ITS VALUES ON A BASIS OF THE TARGET

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Allowed standard prerequisites (editorial).

- Lemma lemma-restrict-basis-equivalence-structures (canonical tag 009R): for sheaves with values in (C, F), morphisms of sheaves agree with morphisms of their restrictions to a basis of the topology, uniquely. cited at proof lines 4496-4498. Passes from the morphism on the basis to a morphism G -> f_*F.
- Lemma lemma-f-map (canonical tag none, chapter not published): maps G -> f_*F of sheaves of sets are the same as f-maps. cited at proof lines 4498-4499. Converts the sheaf morphism into the claimed f-map.
- Lemma lemma-f-map-sets-algebraic-structures (canonical tag 008Q): the underlying-set f-map of a morphism of sheaves of algebraic structures is an f-map of the underlying sheaves of sets and conversely under the adjointness. cited at proof lines 4500-4502 for the algebraic-structure case. Handles the upgrade from sets to sheaves with values in C.

Source reference. Original definition 008J: sources/stacks-chow/tags.txt, source lines 319–319. Cited at statement lines 4478-4481 fixing the notion being constructed; navigation for the claim.

Source reference. Original section 008P: sources/stacks-chow/tags.txt, source lines 325–325. Cited at statement lines 4479-4482 for the discussion of f-maps in the algebraic setting; navigation.

Source reference. Original lemma 009R: sources/stacks-chow/tags.txt, source lines 363–363. Cited at proof lines 4496-4498; load-bearing inside the counted span.

Source reference. Original lemma 008K: sources/stacks-chow/tags.txt, source lines 320–320. Cited at proof lines 4498-4499 to convert the sheaf morphism into an f-map; load-bearing inside the counted span.

Source reference. Original lemma 008Q: sources/stacks-chow/tags.txt, source lines 326–326. Cited at proof lines 4500-4502 for the algebraic-structure case; load-bearing inside the counted span.

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Lemma 0.2. *Let $f : X \to Y$ be a continuous map of topological spaces. Let $(\mathcal{C}, F)$ be a type of algebraic structures. Let $\mathcal{F}$ be a sheaf with values in $\mathcal{C}$ on X. Let $\mathcal{G}$ be a sheaf with values in $\mathcal{C}$ on Y. Let $\mathcal{B}$ be a basis for the topology on Y. Suppose given for every $V \in \mathcal{B}$ a morphism*

$$\varphi_V : \mathcal{G}(V) \longrightarrow \mathcal{F}(f^{-1}V)$$

of $\mathcal{C}$ compatible with restriction mappings. Then there is a unique f-map (see Definition 008J and discussion of f-maps in Section 008P) $\varphi : \mathcal{G} \to \mathcal{F}$ recovering φ_V for $V \in \mathcal{B}$.

Proof. This is trivial because the collection of maps amounts to a morphism between the restrictions of $\mathcal{G}$ and $f_*\mathcal{F}$ to $\mathcal{B}$. By Lemma 009R this is the same as giving a morphism from $\mathcal{G}$ to $f_*\mathcal{F}$, which by Lemma 008K is the same as an f-map. See also Lemma 008Q and the discussion preceding it for how to deal with the case of sheaves of algebraic structures. □